GOBLIN · OVIEDO BIOHACKATHON 2026 · SHORT UPDATE

SSSOM → RDF
Named graphs & RDF 1.2

Rendering ontology mappings as two complementary RDF serializations, served with Oxigraph and browsed with Comunica.

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⚠ First draft / proposal — serializer generated by prompting an LLM (Claude); outputs need validation & some IRIs need minting (see last slide).

← → arrows to navigate

1 What is SSSOM?

SSSOM — *Simple Standard for Sharing Ontological Mappings*. Its native serialization is a **hybrid of TSV and YAML**: a TSV table of mappings, with the set-level metadata (provenance, confidence, the `curie_map`) carried as a **YAML block in the TSV's leading `#` comment lines**.

```
# mapping_set_id: https://w3id.org/commons/ols/mappings/mondo.ols.sssom.tsv  ← YAML header
# mapping_set_confidence: '0.7'
# curie_map:
#   MONDO: http://purl.obolibrary.org/obo/MONDO_
#   DOID:  http://purl.obolibrary.org/obo/DOID_
subject_id     predicate_id        object_id  mapping_justification      ← TSV table
MONDO:0000001  oboInOwl:hasDbXref  DOID:4     semapv:UnspecifiedMatching
```

A streaming serializer (**generated by prompting an LLM — Claude**) parses the YAML header, expands every CURIE to a full IRI, percent-encodes local parts (SMILES/InChI safety), and emits RDF. First draft — see caveats on the last slide.

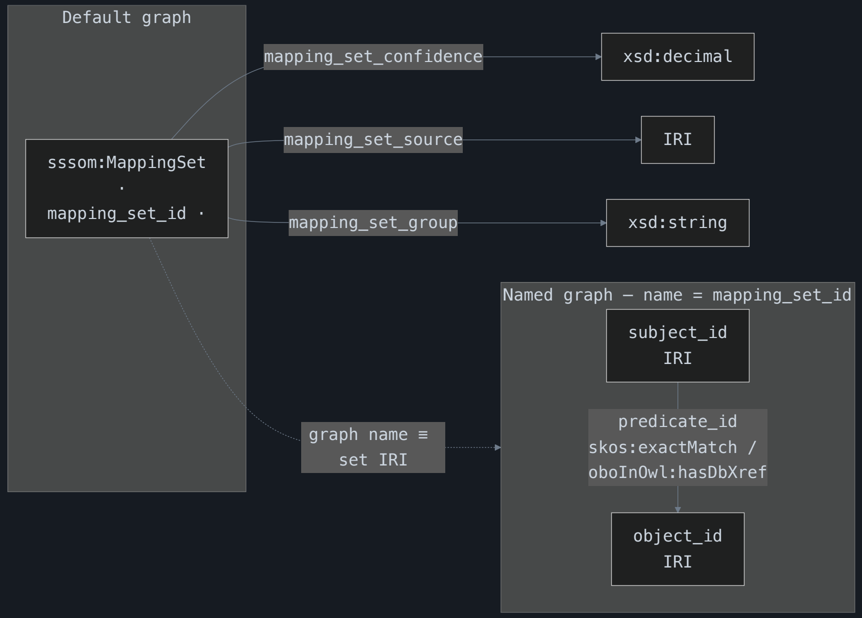
2 Two renderings — the design choice

A mapping is *a statement about a statement*. The two approaches attach that metadata at **different layers of the RDF stack**:

	`Named graphs (TriG)`	`RDF 1.2 (Turtle 1.2)`
Layer	dataset / context → quads	abstract syntax → triple terms
Granularity	one graph per `mapping_set`	per mapping (`{| … |}`)
Carries	set-level provenance	justification + labels (**lossless**)
Semantics	*undefined* by spec (convention)	*defined*; opaque, non-asserting
Tooling	universal — SPARQL 1.1 `GRAPH`	needs SPARQL 1.2 engine (Oxigraph)
Cost	~1 quad / mapping	~5× blow-up (6.2M→34M)
Models	`sssom:MappingSet`	`sssom:Mapping`

Key insight: named graph ↔ `MappingSet`, triple term ↔ `Mapping` — different layers, so they **compose** rather than compete.

3 Schema — named graphs (TriG)



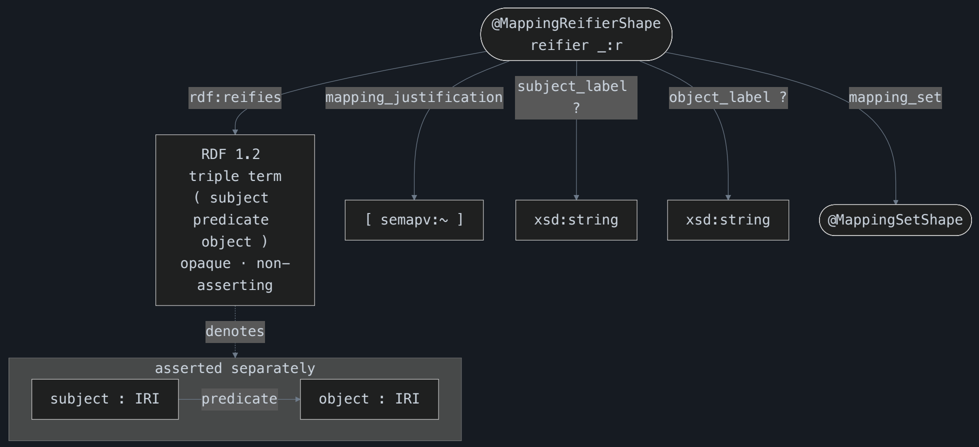
Valid **ShEx** — the per-set resource and the mapped term. (The graph grouping itself is dataset-level, not expressible in ShEx.)

```
PREFIX sssom: <https://w3id.org/sssom/>
PREFIX skos:  <http://www.w3.org/2004/02/skos/core#>

<MappingSetShape> {                    # default graph
  a [ sssom:MappingSet ] ;
  sssom:mapping_set_confidence xsd:decimal ? ;
  sssom:mapping_set_source     IRI ? ;
  sssom:mapping_set_group      xsd:string ?
}
<MappedTermShape> {                     # inside a named graph
  ( skos:exactMatch
  | skos:closeMatch
  | oboInOwl:hasDbXref ) IRI +
}
```

4 Schema — RDF 1.2 (ShEx-style)

⚠ ShEx 2.1 has no triple-term / `rdf:reifies` construct yet – the shape below is **ShEx-style pseudo-syntax**, and the diagram is drawn in ShEx shape style.




```
# pseudo-ShEx (RDF 1.2 not yet in ShEx)
PREFIX sssom:  <https://w3id.org/sssom/>
PREFIX semapv: <https://w3id.org/semapv/vocab/>
PREFIX rdf:    <...22-rdf-syntax-ns#>

<MappingReifierShape> {
  rdf:reifies  TRIPLE_TERM ;           # <<( s p o )>>
  sssom:mapping_justification [ semapv:~ ] ;
  sssom:subject_label  xsd:string ? ;
  sssom:object_label   xsd:string ? ;
  sssom:mapping_set    @<MappingSetShape>
}
```

Each mapping triple is asserted *and* annotated: a reifier links to the opaque triple term and hangs the per-mapping metadata.

5

Sources & status

DONE

EBI OLS

OLS4 publishes per-ontology SSSOM extracts. Full set rendered into both formats.

271	6.24M	~91s
mapping sets	mappings	both formats

TriG 1.5 GB · Turtle 1.2 3.7 GB → in repo via Git LFS (87 / 204 MB gz).

FUTURE WORK

Wikidata

Began extracting ontology cross-references from **Wikidata** as SSSOM; not finished in time.

Comunica **federation** makes the join natural: query local Oxigraph *and* the Wikidata endpoint together.

6 Stack — Oxigraph + Comunica, in the browser

Oxigraph (Rust → **WebAssembly**) parses & queries RDF 1.2 and runs as an in-page triplestore. **Comunica** (browser) fetches & parses the same data and can federate across endpoints. Our demo runs **both engines entirely client-side** over a bundled sample — no server, no endpoint, no download.

▶ Self-contained, in-browser SPARQL playground — pick Oxigraph (WASM) or Comunica, named graphs or RDF 1.2:

Open the live demo →

For the full 5 GB renderings (271 sets / 6.24 M mappings), load them into a real Oxigraph endpoint and query with Comunica — see OXIGRAPH.md.

⚠ Status — a first draft / proposal

This is an early proposal to **start the conversation**, not a finished standard. Being transparent:

- **LLM-generated.** The serializer was produced by **prompting an LLM (Claude)** — a starting point to be audited, not a reference implementation.
- **Validation pending.** Outputs *parse* with a strict RDF 1.2 parser (pyoxigraph), but parsing ≠ correct modelling — the shapes need review against the SSSOM community.
- **IRIs need minting.** `sssom:mapping_set` (a convenience back-link, not a real SSSOM slot) and the `w3id.org/sssom/.well-known/curie/…` fallback are placeholders under `w3id.org` that we don't own. Reused `sssom:` slot IRIs must be verified against the official model.
- **Scope.** No literal-subject mappings (generalized RDF); propagatable slots (license, mapping_date) & extension_definitions not yet emitted by the script.

Full write-up: docs/approach.md

Recap & links

✅ SSSOM (TSV+YAML) → named graphs **and** RDF 1.2, both validated · 271 OLS sets / 6.24M mappings · schemas drawn (ShEx + ShEx-style) · served via Oxigraph, queried via Comunica · 🔭 Wikidata next.

- 🐙 github.com/andrawaag/GOBLIN_Oviedo_hackathon_2026
- 📦 Data (Git LFS): see data/README.md
- ⚙️ OXIGRAPH.md · 🔎 COMUNICA.md
- 📐 Schemas: named graphs · RDF 1.2
- 📝 Approach & caveats: docs/approach.md

Andra Waagmeester (Amsterdam UMC) · Jerven Bolleman (SIB) · Ruben Taelman (IDLab, Ghent University) · Thomas Pellissier Tanon — first draft, LLM-assisted, for discussion.